Dental Sleep Medicine & Sleep Surgery Orthodontic Perspective

Audrey Yoon DDS MS

Clinical Professor

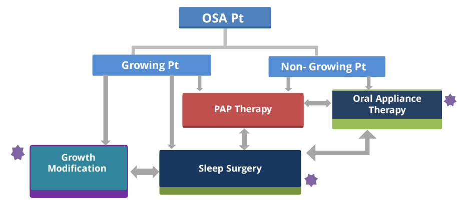
Stanford University, Sleep Medicine Clinic

Diplomate, American Board of Orthodontics

Diplomate, American Board of Dental Sleep Medicine

1

Interdisciplinary OSA Treatment Protocol



2

Treatment Options: CPAP

- Continuous Positive Airway Pressure (CPAP) uses a mask and hose to blow pressurized air into the airway to keep it open during sleep

3

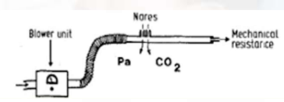
The Original Invention of CPAP

Collin Sullivan 1981

REVERSAL OF OBSTRUCTIVE SLEEP APNOEA BY CONTINUOUS POSITIVE AIRWAY PRESSURE APPLIED THROUGH THE NARES

COLIN E. SULLIVAN, FAIQ G. ISSA, MICHAEL BERTHON-JONES, LORRAINE EVES

Department of Medicine, University of Sydney, New South Wales 2006, Australia



"We actually made the first blowers using a **two-stage vacuum cleaner motor** that we modified for home use. We tried a whole range of head harnesses, but one of the earliest setups that worked was using the **inside of a bicycle helmet to hold the mask in place**. We also **individually molded masks and provided patients with large pots of Silastic paste and catalyst to glue the masks on.** We later accessed the Vortex blower motor, which was designed for dental drills, and these worked really well for home use and were easily accessible. "

Sullivan et al. Lancet 1981; 1:862

4

CPAP is an amazing treatment

But maybe not the sexiest treatment

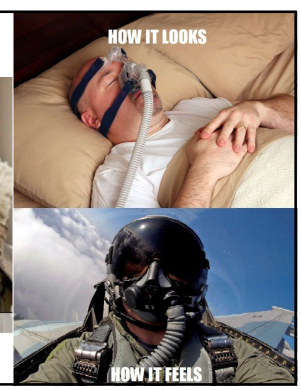
The Ultimate Surfer Chick

Mr. "Shirts are Always Optional"

Audrey Yoon, DDS, MS

5

It can be scary and it can be uncomfortable



Audrey Yoon, DDS, MS

6

Issues with CPAP Non-Compliance

- 50% discontinue therapy within the first weeks Overall compliance drops to 30% to 60% after several years.
- Untreated OSA patients utilized up to 50% more medical resources and accumulated more than double the healthcare costs.
- Weaver et al. Sleep 2007:
  - 4h use/night for improvement in the Sleepiness Scale (ESS) to less than 11

Audrey Yoon, DDS, MS

7

Oral Appliance
(Mandibular Advancement Device)

8

Oral Appliances

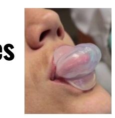
Tongue-Retaining Devices (TRDs)

Mandibular Advancement Devices (MADs)

Audrey Yoon, DDS, MS

9

Tongue Retaining Devices



- Samelson and Cartwright 1982
- Aimed for his own chronic snoring

-> hold the tongue in the most protruded position

Audrey Yoon, DDS, MS

10

Mandibular Advancement Devices

Pierre Robin

- Underdevelopment of the mandible
- Retraction of the tongue(Glossoptosis) .
- "Monobloc" first described in 1902

Monoblock appliance

Robin P. Observation sur un nouvel appareil de redressement. Rev Stomatol 1902; 9:423

Audrey Yoon, DDS, MS

11

Patient Selection
Devices features
Provider Expertise

Device Features and Design

Adjustable? Comfortable? Durable? Long Term Side-effect?

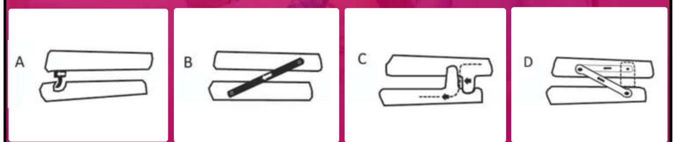
Dorsal Fin | Herbst | Cad Cam | Airway | Facial Integrity

Audrey Yoon, DDS, MS

12

Device Features and Design

Patient Selection
Devices features
Provider Expertise



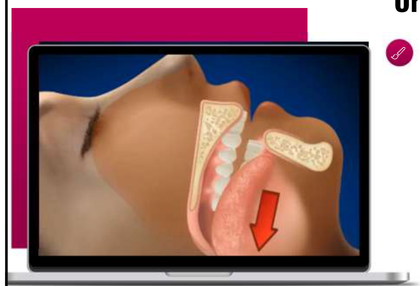
MIDLINE ANTERIOR PUSH/PULL | BILATERAL PUSH | BILATERAL INTERLOCK OR FASTENED | BILATERAL PULL

Audrey Yoon, DDS, MS

Essick et al. Oral appliance therapy for sleep-disordered breathing part III. Inside dental technology. September 2016, 28-32.

13

Oral Appliances



Oral appliances are **custom-made** mouthpieces intended to pull the jaw forward during sleep to help open upper airway

Audrey Yoon, DDS, MS

14

Oral Appliances

Good **alternative** to CPAP
- Primary snoring
- Mild to moderate OSA
- Severe OSA in case of CPAP intolerance

*The **American Academy of Sleep Medicine (1995)***

An effective **first-line** treatment
- Primary Snoring
- Mild to moderate OSA

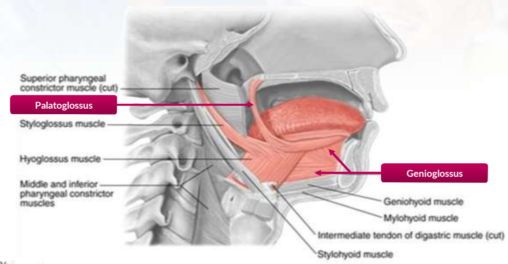
*The **American Academy of Sleep Medicine (2006)***

Audrey Yoon, DDS, MS

15

How Does Oral Appliance Work ?

(Kim et al, Eur Arch Otorhinolaryngol 2018; Thorax , Ryan Et Al 1999)

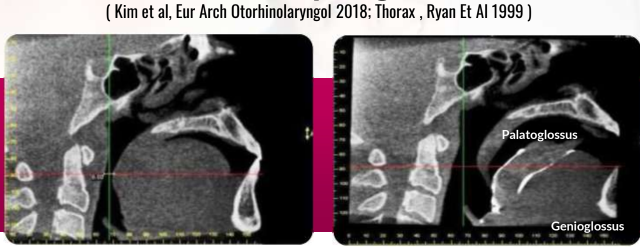


Audrey Yoon, DDS, MS

16

Mechanism: 3D Opening Of Pharynx

(Kim et al, Eur Arch Otorhinolaryngol 2018; Thorax , Ryan Et Al 1999)



Oral appliances **pull tongue and hyoid** and **upright soft palate**
-> Retroglossal and **Retropalatal opening**

Audrey Yoon, DDS, MS

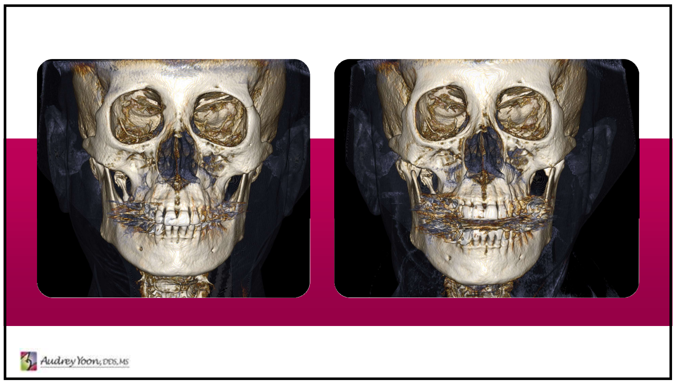
17

Expanding Lateral Pharyngeal Walls

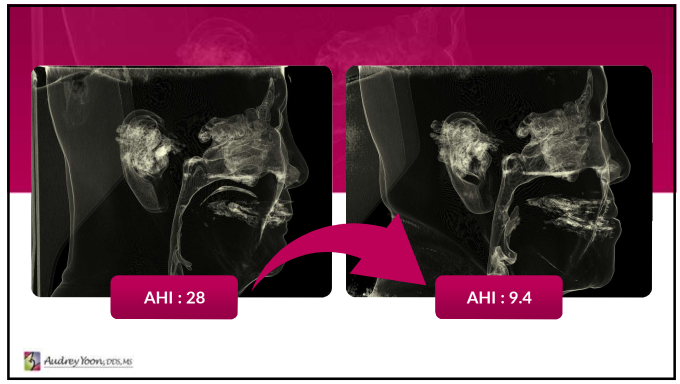
(Thorax , Ryan Et Al 1999)

Audrey Yoon, DDS, MS

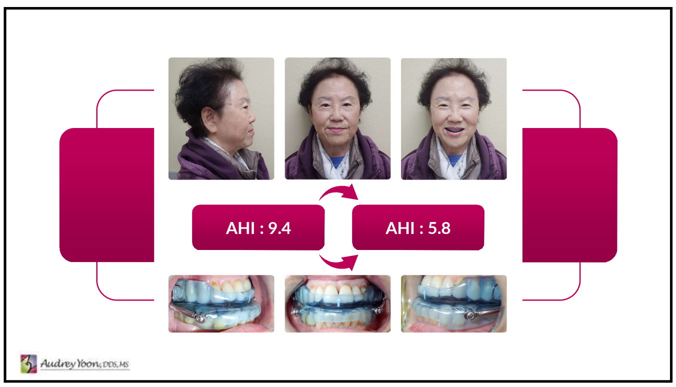
18



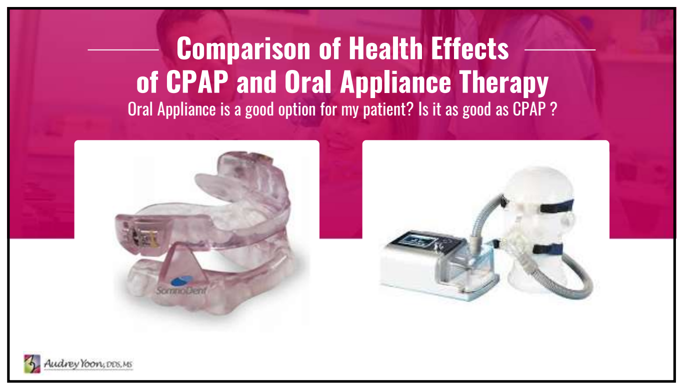
19



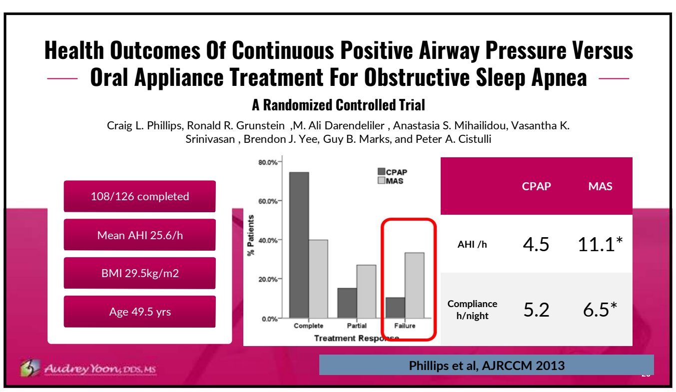
20



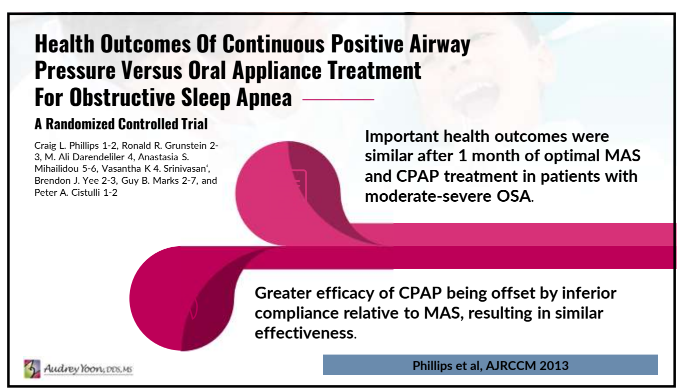
21



22



23



24

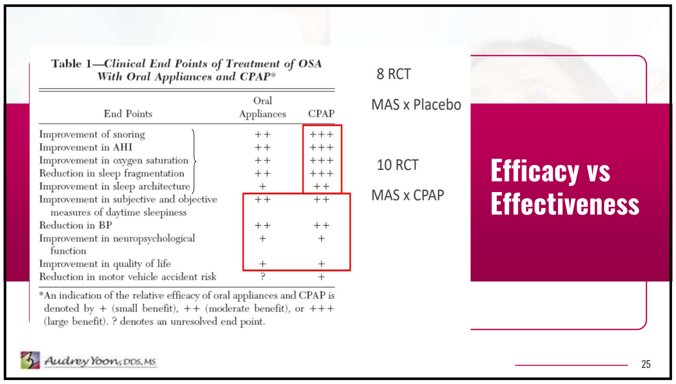
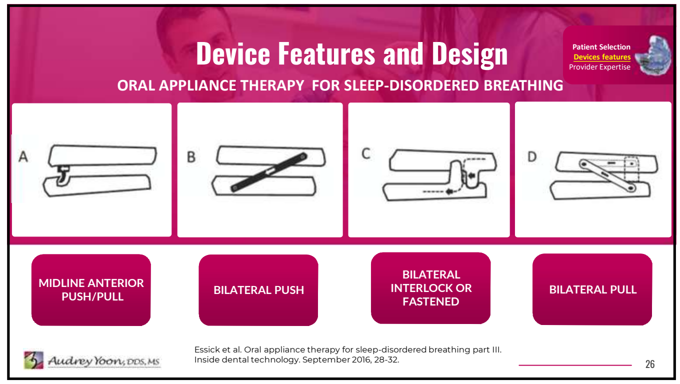


Table 1—*Clinical End Points of Treatment of OSA With Oral Appliances and CPAP**

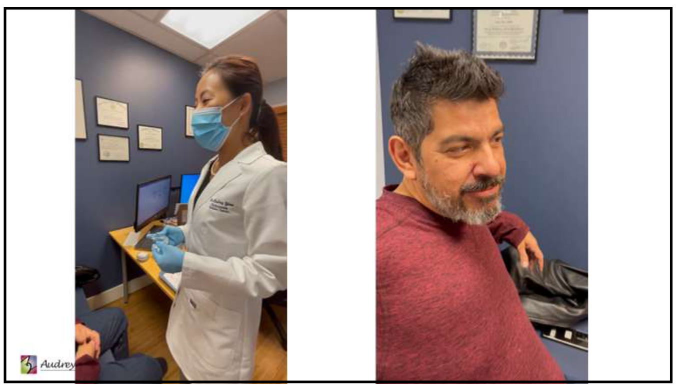
End Points	Oral Appliances	CPAP
Improvement of snoring	++	+++
Improvement in AHI	++	+++
Improvement in oxygen saturation	++	+++
Reduction in sleep fragmentation	++	+++
Improvement in sleep architecture	+	++
Improvement in subjective and objective measures of daytime sleepiness	++	++
Reduction in BP	++	++
Improvement in neuropsychological function	+	+
Improvement in quality of life	+	+
Reduction in motor vehicle accident risk	?	+

*An indication of the relative efficacy of oral appliances and CPAP is denoted by + (small benefit), ++ (moderate benefit), or +++ (large benefit). ? denotes an unresolved end point.

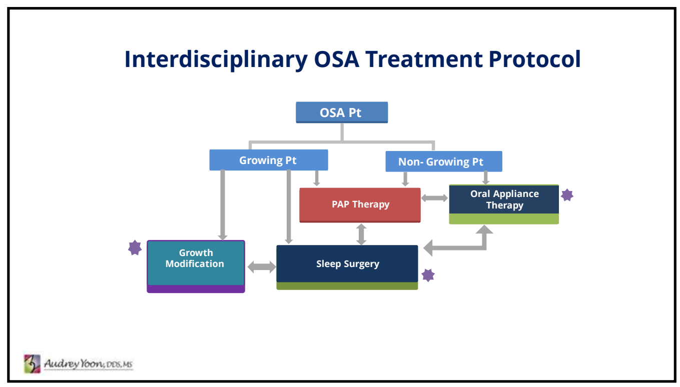
25



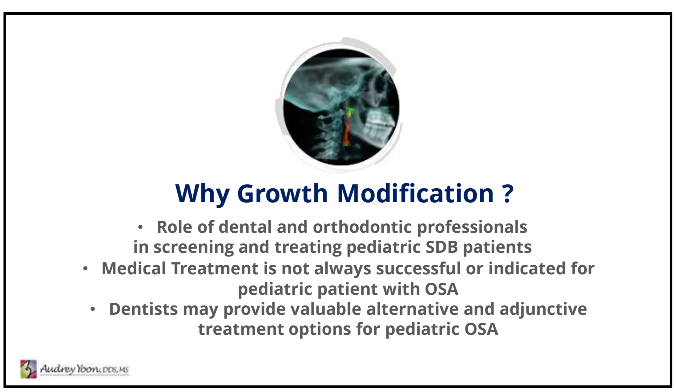
26



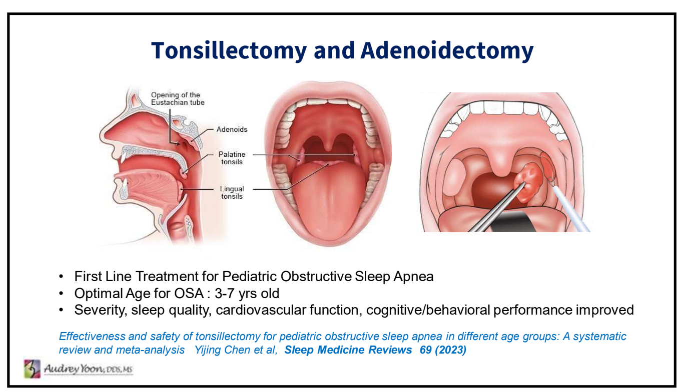
27



28



29



30

31

Sleep Medicine 66 (2020) 282–283

Contents lists available at ScienceDirect

Sleep Medicine

journal homepage: www.elsevier.com/locate/sleep

Editorial

Pediatric obstructive sleep apnea: beyond adenotonsillectomy

Almiro J. Machado Júnior*, Agrício N. Crespo
Department of Otorhinolaryngology, Head and Neck Surgery, University of Campinas (UNICAMP), Campinas, São Paulo, Brazil

and, consequently, leading to airway collapse. At present, the treatment of choice worldwide for OSAS during childhood is adenotonsillectomy (AT). AT procedures account for 15% of all pediatric surgery carried out in the United States [2].

However, newer studies have shown that even after AT, 60%–70% of children experience residual OSAS. A recent meta-analysis showed that the short-term results from AT were divergent, and that AT did not provide any significant long-term benefit [3,4].

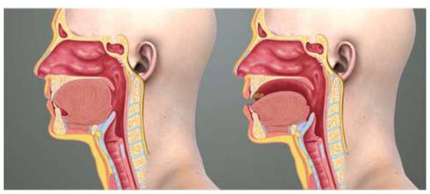
Therefore, the following minimally invasive coadjuvant treatments have recently been proposed: (1) rapid maxillary expansion (RME), (2) use of oral appliances (OA) and (3) myofunctional therapy (MT) [7,10,12–14].

In conclusion, OSAS during childhood leads to physical and neuropsychomotor impairment. As such, it needs to be recognized and treated early, in order to avoid or mitigate its deleterious consequences and allow children to develop properly. Thus, it becomes the task of healthcare professionals to research and disseminate new forms of therapy for treating childhood OSAS.

Audrey Yoon, DDS, MS

32

Nasal Breathing vs Mouth Breathing



- Nasal Passage for filtering, warming and humidifying the inspired air
- Increase oxygen uptake and circulation
- Facilitates inhalation of Nitric Oxide and prevent infections and allergic reaction & helps in boosting immunity

- Foreign antigens presented to immune cells of tonsils through mouth
-> trigger an immune response and tonsillar hypertrophy

Audrey Yoon, DDS, MS

33

Nasal Breathing and Nitric Oxide

Nitric oxide (NO)

- One of the most important signal molecules in the body : selectively mediating cell signaling
- Keep blood vessels flexible, Encourage vasodilation
- Lower blood pressure
- Improve mood & cognition
- Plays a role in maintaining upper airway dilator muscles tone, regulation of spontaneous respiration, and neuromuscular control during sleep.
- ***Helps in boosting immunity & rids of bacteria in airborne particles***
- ***Produced in the nose and the paranasal sinuses : During Nasal Breathing***

Nitric Oxide (NO)

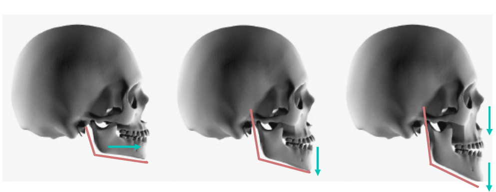
The Nobel Prize in Physiology or Medicine 1998

Haight JS, Djupesland PG. Nitric oxide (NO) and obstructive sleep apnea (OSA). *Sleep Breath.* Jun 2003

Audrey Yoon, DDS, MS

34

Vertical growth pattern : Mouth Breather (Clockwise growth pattern)



Tongue push forward

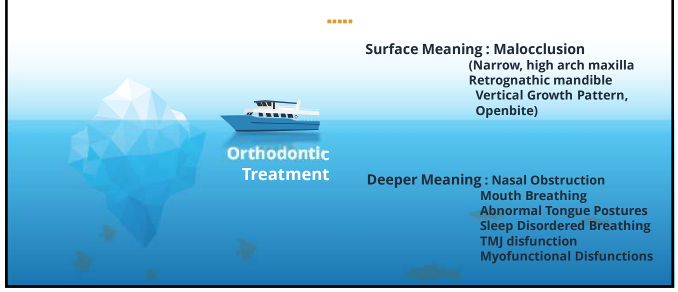
Mouth Breathing : Mouth open

Open Bite
Long Face
Vertical Growth

Audrey Yoon, DDS, MS

35

Malocclusion can be a sign of other medical conditions

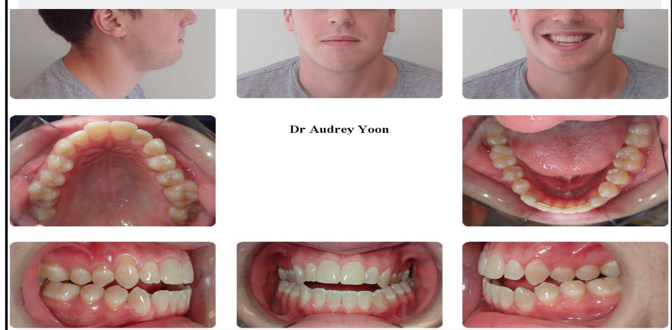


36

Understanding Malocclusion

Airway affects development of malocclusion

37

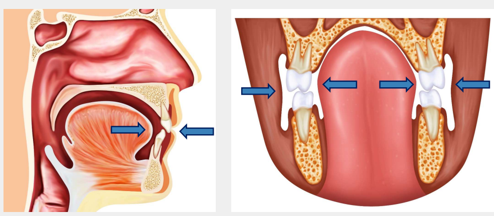


38

Zone of Equilibrium : Neutral Zone

The forces of tongue pressing outward are neutralized by the forces of the cheeks and lips

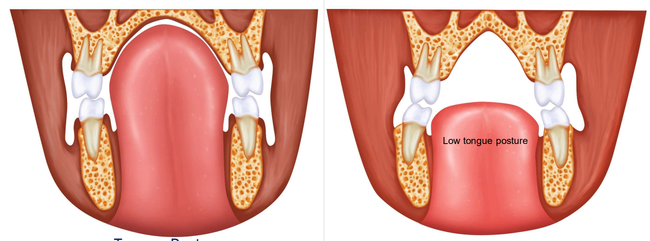
Chewing, speaking, swallowing and Breathing



Marc Geissberger. *Esthetic Dentistry in Clinical Practice*. John Wiley & Sons, 2013. ISBN 9781118694930.

39

Zone of Equilibrium : Neutral Zone



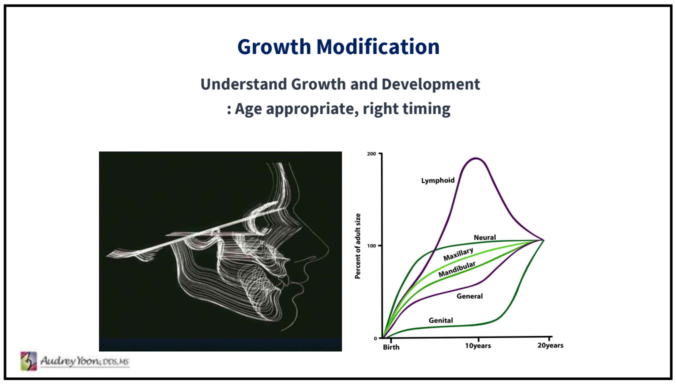
Tongue Posture

Relates to Habits, Tongue Tie, Mouth breathing, Nasal Blockage, Adenoids and tonsils, Allergies and Asthma ⟷ Narrow, high arch palate

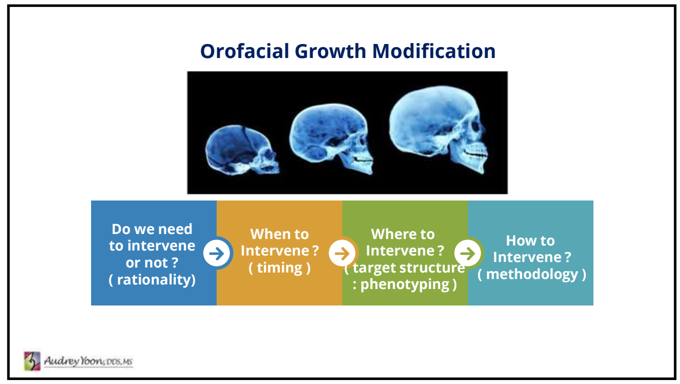
40

41

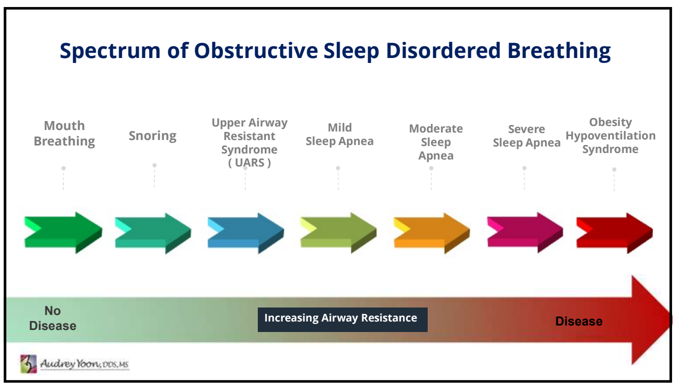
42



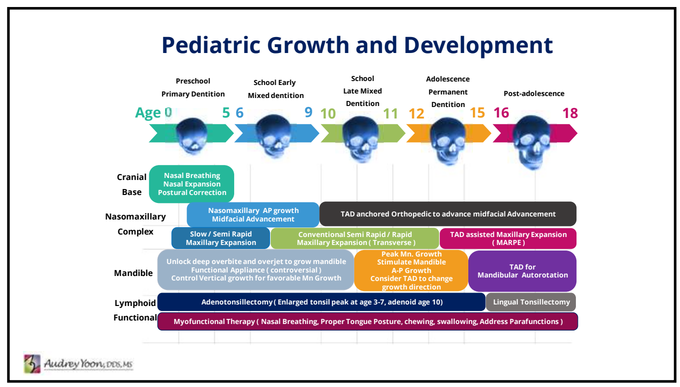
43



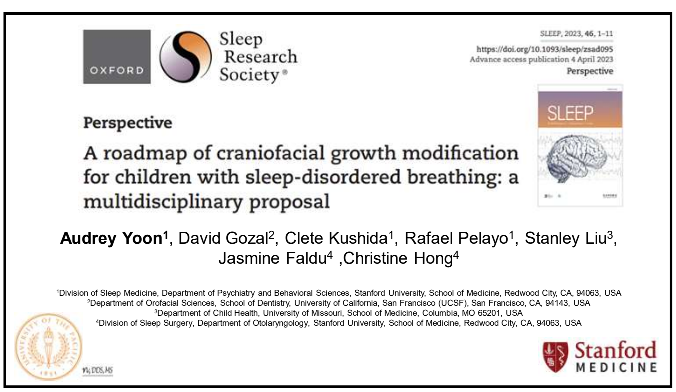
44



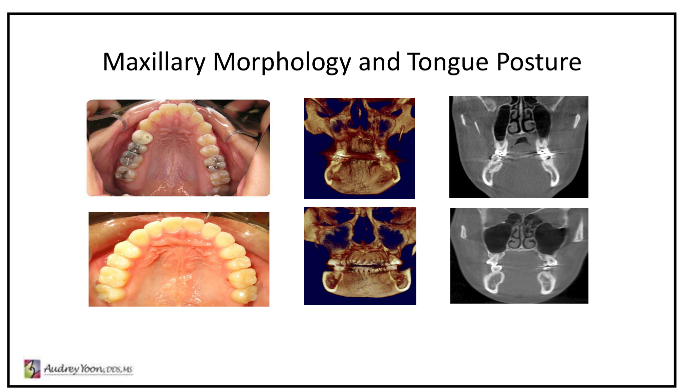
45



46

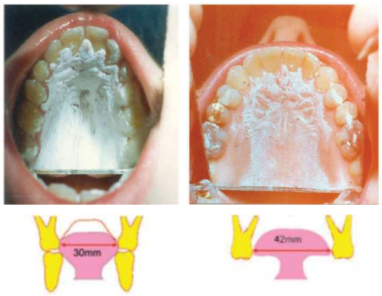


47



48

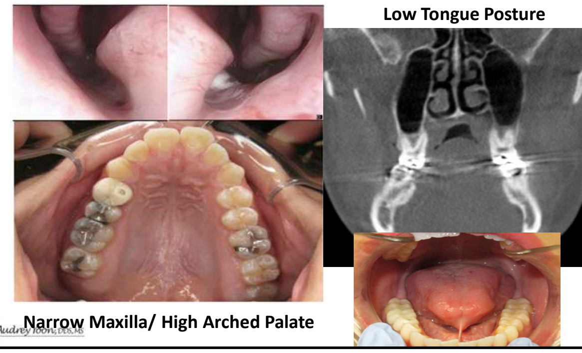
Tongue Posture



49

Narrow Nasal Floor

Low Tongue Posture



Narrow Maxilla/ High Arched Palate

50

Treatment of Maxillary Transverse Deficiency
→ Rapid Palatal/Maxillary Expander (RPE / RME)

51

Landmark Paper : Sleep 2004 Pirelli et al

Rapid Maxillary Expansion in Children with Obstructive Sleep Apnea Syndrome

Paola Pirelli, DDS[1]; Maurizio Saponara, MD[2]; Christian Guilleminault, MD, Biol D[3]

[1]Department of Odontological Sciences, University Tor Vergata; [2]Department of Neurology and ENT, La Sapienza University, Rome Italy; [3]Stanford University Sleep Disorders Clinic, Stanford, CA, USA

- 31 children, mean age 8.7, no adenotonsillar hypertrophy, upper jaw constriction
- AHI 12.2 per hour → AHI <1 after 4 months (all of children)

Follow up Paper : Sleep medicine 2015 Pirelli et al

Rapid maxillary expansion (RME) for pediatric obstructive sleep apnea: a 12-year follow-up

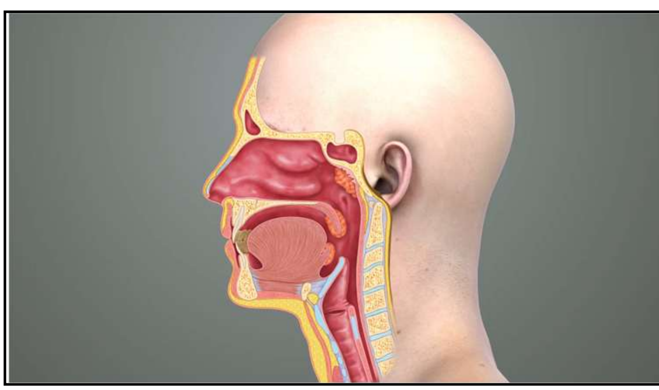
Paola Pirelli [a], Maurizio Saponara [b], Christian Guilleminault [c,*]

- 31 children, yearly check up for 12 yrs for 23 children (73 % initial group)
- Long term, stable post RME treatment (PSG normal, expansion stable)

52

How Does Palatal Expansion Improve Sleep Disordered Breathing ?

53



54

How Does Palatal Expansions IMPROVE OSA?

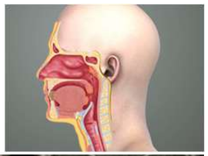
1 . Nasopharynx

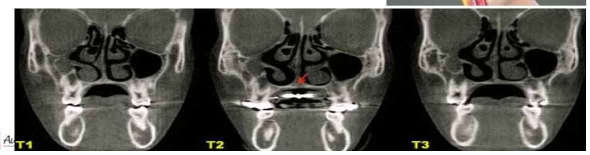
: Expand nasal cavity

→ Decrease resistance to airflow through the nose

→ Improve nasal breathing

(Deeb at al. Am J Orthod Dentofacial Orthop 2010)
(Zeng at al. Sleep 2008)





55

How Does Palatal Expansions IMPROVE OSA?

1 . Nasopharynx

: Expand nasal cavity

→ Expand nasal cavity volume

→ Expand nasal floor

→ Decrease resistance to airflow through the nose

→ Improve nasal breathing

(Deeb at al. Am J Orthod Dentofacial Orthop 2010)
(Zeng at al. Sleep 2008)

Audrey Yoon, DDS,MS

56

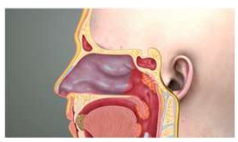
How Does Palatal Expansions IMPROVE OSA?

2. Oral Cavity : Expand oral cavity

→ Create more tongue spaces

(esp. roof of the mouth)

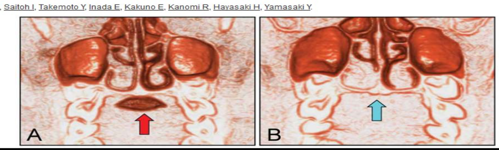
→ Improve tongue posture



Am J Orthod Dentofacial Orthop. 2013 Feb;143(2):235-45. doi: 10.1016/j.ajodo.2012.09.014.

Tongue posture improvement and pharyngeal airway enlargement as secondary effects of rapid maxillary expansion: a cone-beam computed tomography study.

Iwasaki T[1], Saitoh I, Takemoto Y, Inada E, Kakuno E, Kanomi R, Hayasaki H, Yamasaki Y.



57

How Does Palatal Expansions IMPROVE OSA?

2. Oral Cavity : Expand oral cavity

→ Improve tongue posture (Advance and raise the tongue into the oral cavity) → Expand posterior airway space

Am J Orthod Dentofacial Orthop. 2013 Feb;143(2):235-45. doi: 10.1016/j.ajodo.2012.09.014.

Tongue posture improvement and pharyngeal airway enlargement as secondary effects of rapid maxillary expansion: a cone-beam computed tomography study.

Iwasaki T[1], Saitoh I, Takemoto Y, Inada E, Kakuno E, Kanomi R, Hayasaki H, Yamasaki Y

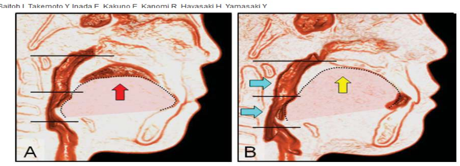


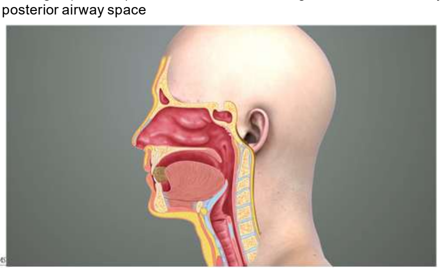
Fig 7. Improvement of low tongue posture and enlargement of the pharyngeal airway after RME in a patient: **A,** before RME, tongue posture is low (*red arrow*), and the oropharyngeal airway is narrow; **B,** after RME, tongue posture has improved (*yellow arrow*), and the pharyngeal airway has enlarged (*blue arrows*).

58

How Does Palatal Expansions IMPROVE OSA?

2. Oral Cavity : Expand oral cavity

→ Improve tongue posture (Advance and raise the tongue into the oral cavity) → Expand posterior airway space

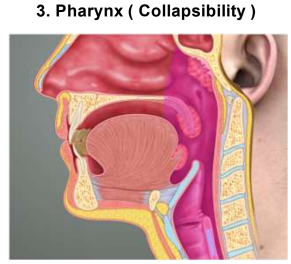


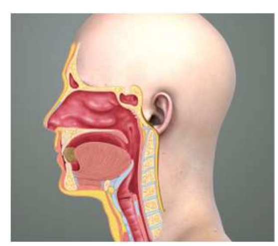
Audrey Yoon, DDS,MS

59

How Does Palatal Expansions IMPROVE OSA?

3. Pharynx (Collapsibility)





Audrey Yoon, DDS,MS

60

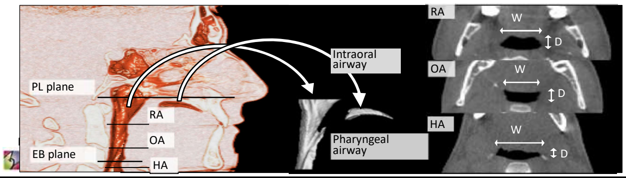
Sleep and Breathing
https://doi.org/10.1007/s11325-019-01948-7
ENT • ORIGINAL ARTICLE

Check for updates

How does distraction osteogenesis maxillary expansion (DOME) reduce severity of obstructive sleep apnea?

Tomonori Iwasaki[1] • Audrey Yoon[2] • Christian Guilleminault[3] • Youichi Yamasaki[1] • Stanley Yung Liu[4]

Received: 27 December 2018 / Revised: 30 August 2019 / Accepted: 18 September 2019

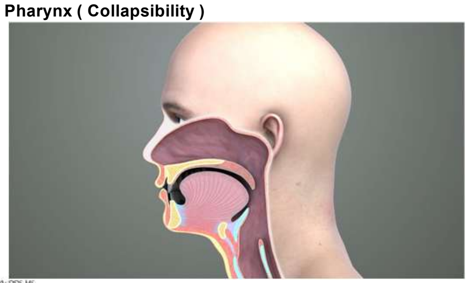


Audrey Yoon, DDS, MS

61

How Does Palatal Expansions IMPROVE OSA?

3. Pharynx (Collapsibility)

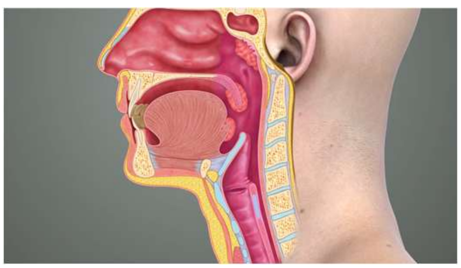


Audrey Yoon, DDS, MS

62

How Does Palatal Expansions IMPROVE OSA?

4. Tonsils and Adenoids



Audrey Yoon, DDS, MS

63

ELSEVIER
Sleep Medicine
Volume 92, April 2022, Pages 96-102
Original Article

Impact of rapid palatal expansion on the size of adenoids and tonsils in children

Audrey Yoon[a], Mohamed Abdelwahab[b], Rebecca Bockow[c], Ava Vakili[d], Katherine Lovell[d], Inwon Chang[e], Rumpa Ganguly[f], Stanley Liu[b], Clete Kushida[a], Christine Hong[g]

Audrey Yoon, DDS, MS

64

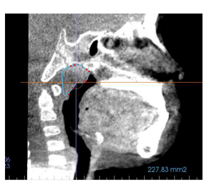
Sleep Medicine
Volume 92, April 2022, Pages 96-102
Original Article
Impact of rapid palatal expansion on the size of adenoids and tonsils in children
Audrey Yoon[a], Mohamed Abdelwahab[b], Rebecca Bockow[c], Ava Vakili[d], Katherine Lovell[d], Inwon Chang[e], Rumpa Ganguly[f], Stanley Liu[b], Clete Kushida[a], Christine Hong[g]

Objectives

- To compare and evaluate change in pts' adenoid and palatine tonsils with RPE using 3D volumetric analysis

Materials and Methods

- Total 60 pts: 40 RPE group vs 20 Control group (no tx)
- Mean ages: 9.79 yrs old, range 5-15, 32 females, 28 males
- CBCT acquired at Initial & 13 months after RPE
- Measures in difference of adenoid and tonsil volume.



Audrey Yoon, DDS, MS

65

65

Sleep Medicine
Volume 92, April 2022, Pages 96-102
Original Article
Impact of rapid palatal expansion on the size of adenoids and tonsils in children
Audrey Yoon[a], Mohamed Abdelwahab[b], Rebecca Bockow[c], Ava Vakili[d], Katherine Lovell[d], Inwon Chang[e], Rumpa Ganguly[f], Stanley Liu[b], Clete Kushida[a], Christine Hong[g]

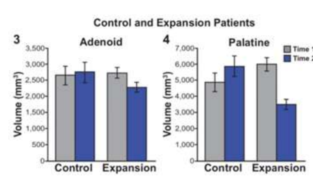
Results

- RPE group: 97.5 % of pt tonsil shrinkage, 90% of pt adenoid reduction
- Average volume reduction

: Adenoids 16.8 % and Tonsils 38.5%

Conclusion

- RPE significantly reduced the size of both adenoid and palatine tonsils **: another long-term benefit of RPE treatment**



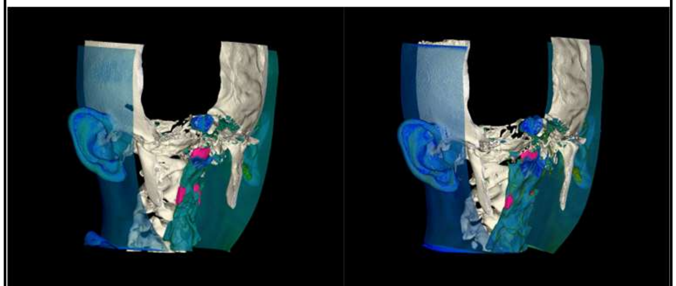
Audrey Yoon, DDS, MS

66

66

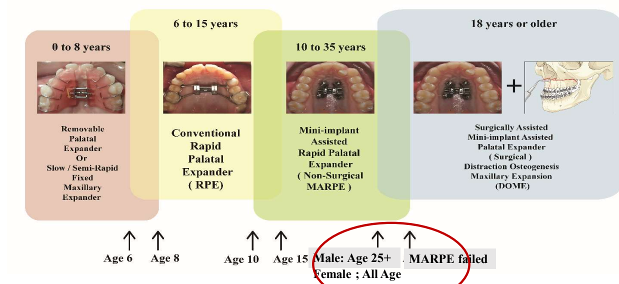
Videos (Auto-play)

Before RPE | After RPE



67

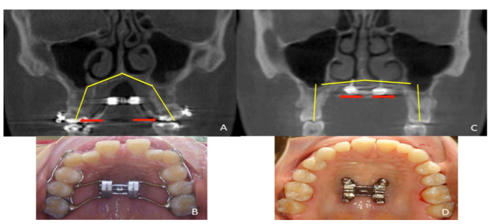
Maxillary Expansion Protocol for OSA



68

Maxillary Expansion using TAD

Dental Expansion | Skeletal Expansion



Audrey Yoon, DDS,MS

69

Brunetto et al. Progress in Orthodontics (2022) 23:3
https://doi.org/10.1186/s40510-021-00397-x

Progress in Orthodontics

RESEARCH | Open Access

Mini-implant assisted rapid palatal expansion (MARPE) effects on adult obstructive sleep apnea (OSA) and quality of life: a multi-center prospective controlled trial

Daniel Paludo Brunetto, Christoph E. Moschik, Ramon Dominguez-Mompell, Eliza Jaria, Eduardo Franzotti Sant'Anna and Won Moon

MARPE (N=14) : AHI 28.8→ AHI 11.4
Control (N=10) : AHI 27.4→ AHI 24.2

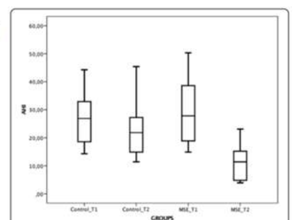


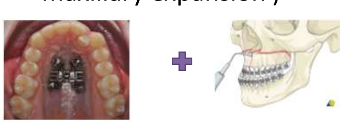
Fig. 4 Boxplot comparing the AHI among the groups and timepoints

Audrey Yoon, DDS,MS

70

MARPE (miniscrew assisted rapid palatal expander)

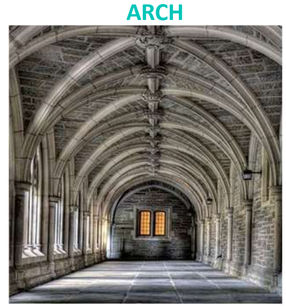
DOME (distraction osteogenesis maxillary expansion)

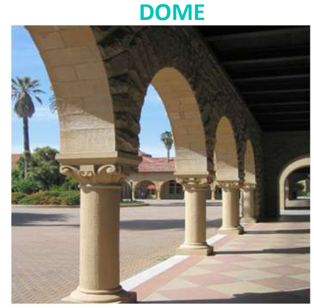


Audrey Yoon, DDS,MS

71

ARCH | **DOME**



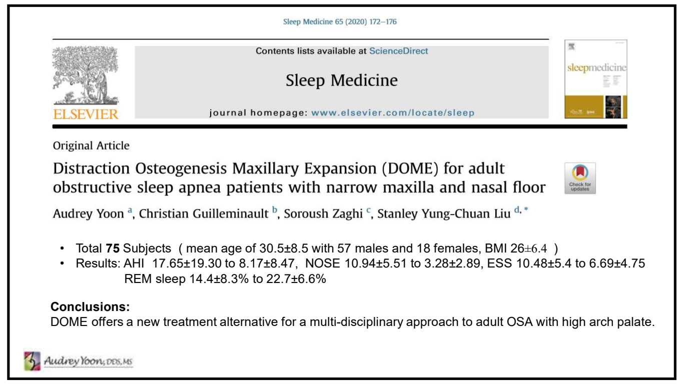


Distraction Osteogenesis for Maxillary Expansion

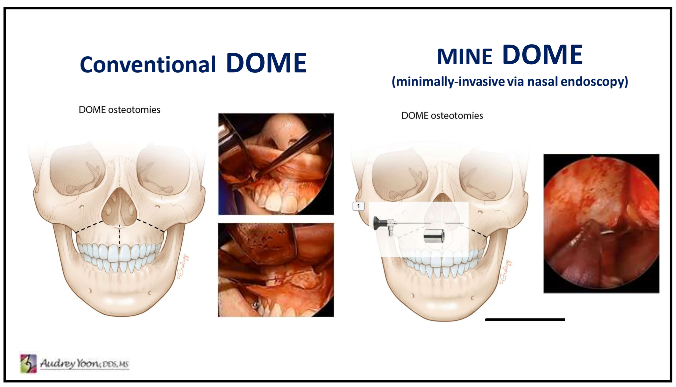
Audrey Yoon, DDS,MS

72

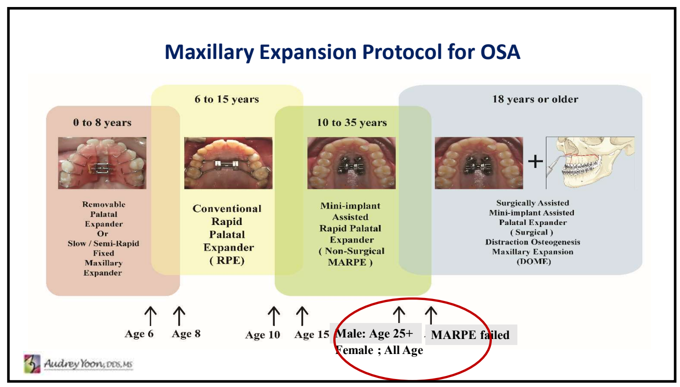
73



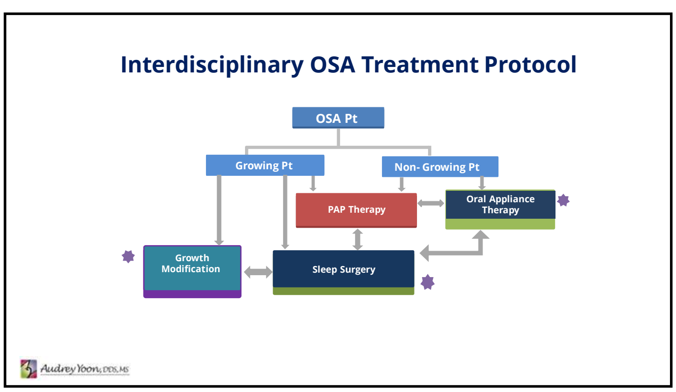
74



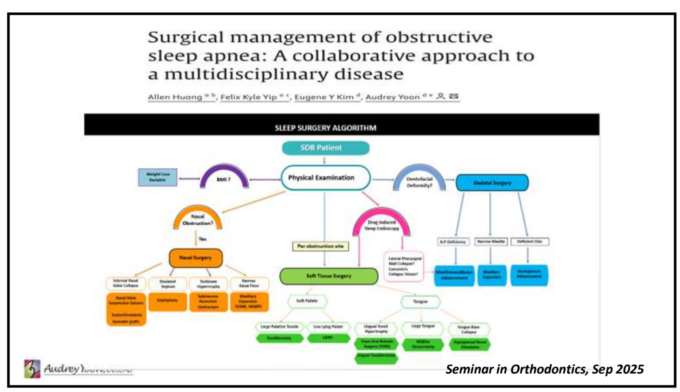
75



76



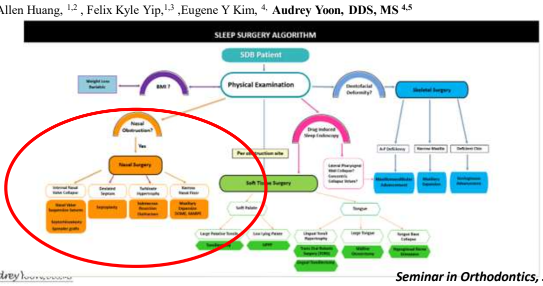
77



78

Surgical Management of Obstructive Sleep Apnea: A Collaborative Approach to a Multidisciplinary Disease

Allen Huang, [1,2], Felix Kyle Yip,[1,3], Eugene Y Kim, [4], **Audrey Yoon, DDS, MS** [4,5]

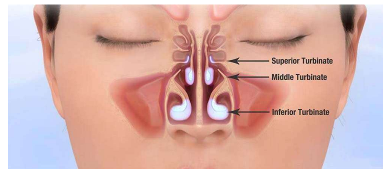


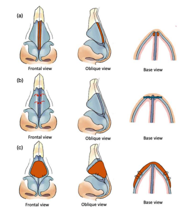
Audrey Yoon, DDS, MS

Seminar in Orthodontics, Sep 2025

79

Nasal Surgery





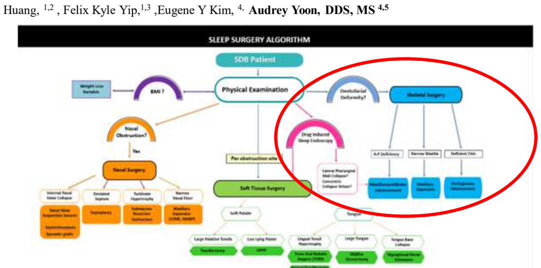
1. **Types of Nasal Surgeries:**
   1. **Septoplasty** – corrects deviated septum
   2. **Inferior Turbinate Reduction** – reduces swollen tissue
   3. **Nasal Valve Repair** – addresses internal/external collapse
2. **Goal of Surgery:**
   1. Improve airflow
   2. Enhance CPAP tolerance
   3. May be combined with other procedures

Audrey Yoon, DDS, MS

80

Surgical Management of Obstructive Sleep Apnea: A Collaborative Approach to a Multidisciplinary Disease

Allen Huang, [1,2], Felix Kyle Yip,[1,3], Eugene Y Kim, [4], **Audrey Yoon, DDS, MS** [4,5]



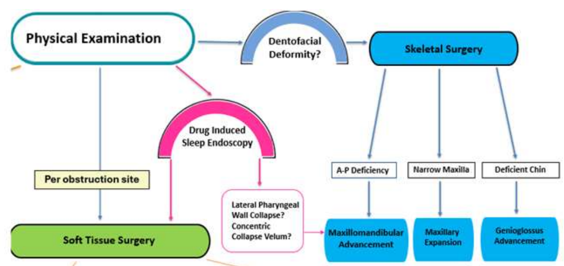
Audrey Yoon

Seminar in Orthodontics, In press 2025

81

Surgical Management of Obstructive Sleep Apnea: A Collaborative Approach to a Multidisciplinary Disease

Allen Huang, [1,2], Felix Kyle Yip,[1,3], Eugene Y Kim, [4], **Audrey Yoon, DDS, MS** [4,5]



Audrey Yoon, DDS, MS

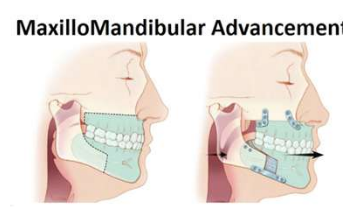
82

MaxilloMandibular Advancement (MMA)

83

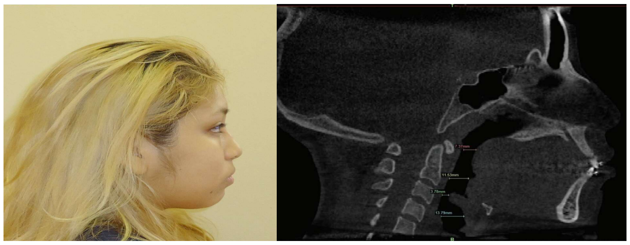
Classic Maxillomandibular Advancement Surgery for OSA

- Developed by Robert Riley, MD, DDS; Nelson Powell, MD, DDS; and Christian Guilleminault, MD at Stanford
- **Maxillary driven procedure** – greater success when maxilla is advanced 10 mm or more and mandible over 11 mm
- Designed to maximize soft tissue tension - allow for larger advancements.

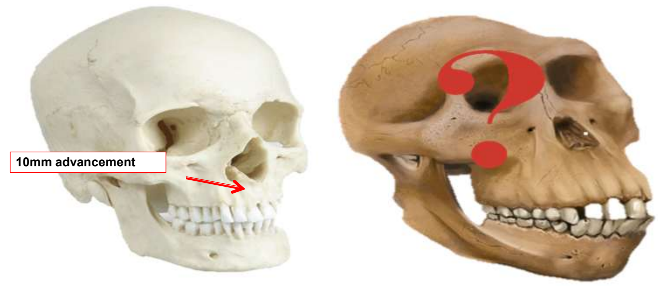


84

MMA for Bi Max Case

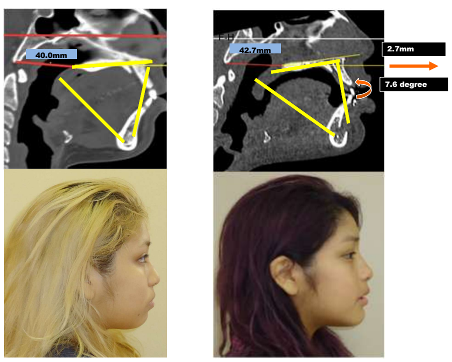


85



Slide courtesy of Dr. Stanley Liu

86



87


Otolaryngology–
Head and Neck Surgery

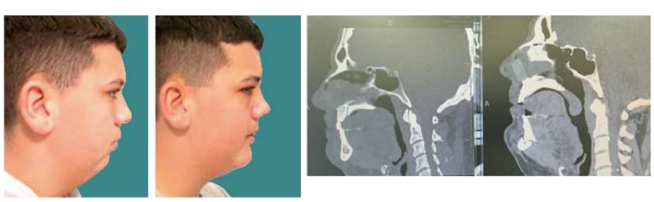
An Accurate Method of Designing and Performing Individual-Specific Genioglossus Advancement

Stanley Yung-Chuan Liu, MD, DDS[1], Leh-Kiong Huon, MD[1,2,3], Soroush Zaghi, MD[1], Robert Riley, MD, DDS[1], and Carlos Torre, MD[1]

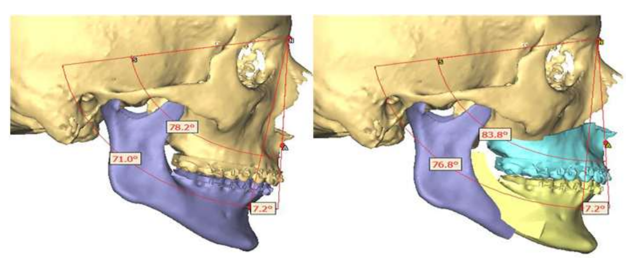
©SAGE

88

Genioplasty + Genioglossal Advancement

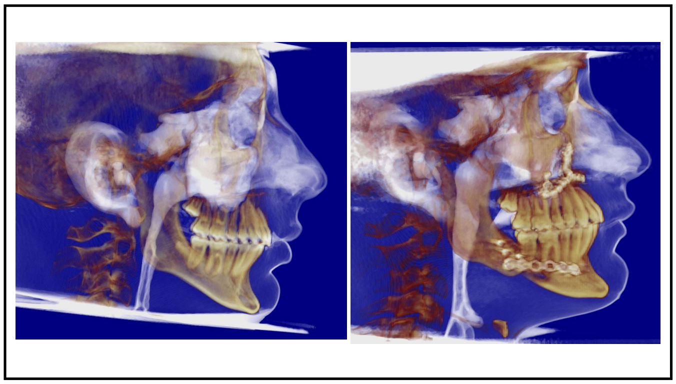


89

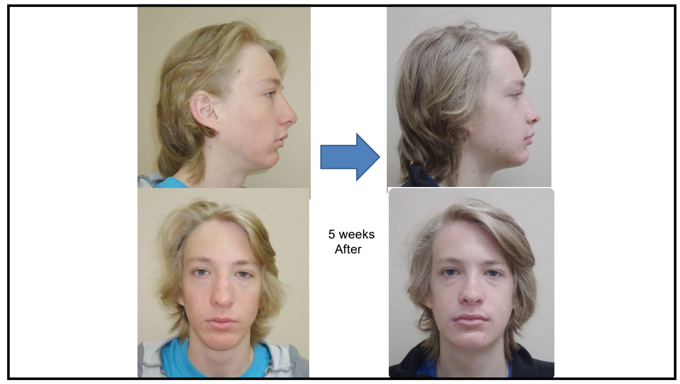


Stanford MEDICINE | Otolaryngology Head & Neck Surgery

90

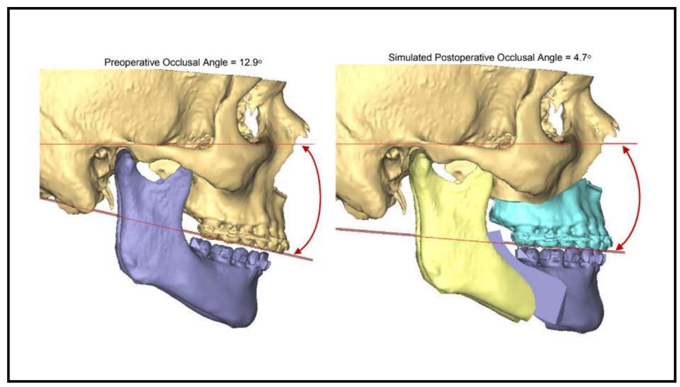


91

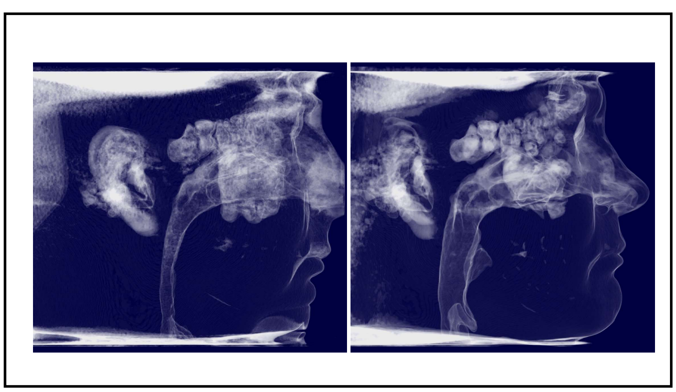


92

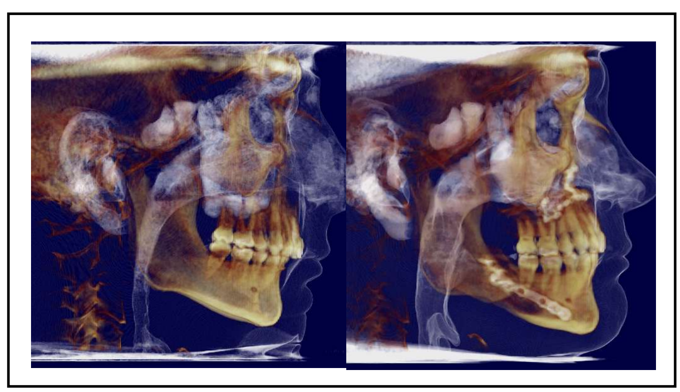
93



94



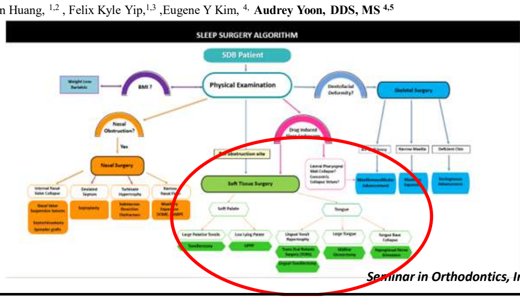
95



96

Surgical Management of Obstructive Sleep Apnea: A Collaborative Approach to a Multidisciplinary Disease

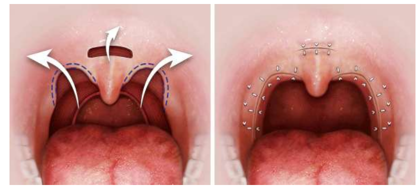
Allen Huang, [1,2] , Felix Kyle Yip,[1,3] ,Eugene Y Kim, [4,] **Audrey Yoon, DDS, MS** [4,5]



Seminar in Orthodontics, In press 2025

97

Palate Surgery



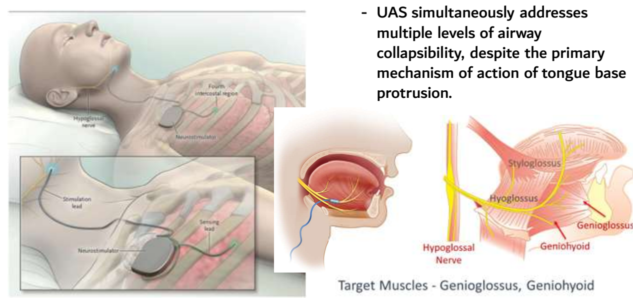
1. Uvulopalatopharyngoplasty (UPPP)
Removes soft palate, uvula, and sometimes tonsils and Widens the oropharyngeal space
2. Other Variants : Expansion pharyngoplasty , Barbed reposition pharyngoplasty (less pain, newer method)
3. Goal of Surgery:
•Reduce vibration (snoring)
•Keep the throat open during sleep
•Often used for mild to moderate OSA

98

Hypoglossal Nerve Stimulation with Maxillary Expansion

99

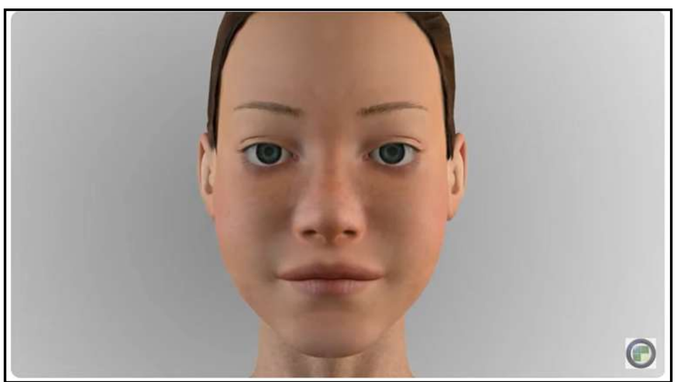
Upper Airway Stimulation (Hypoglossal Nerve Stimulation)



- UAS simultaneously addresses multiple levels of airway collapsibility, despite the primary mechanism of action of tongue base protrusion.

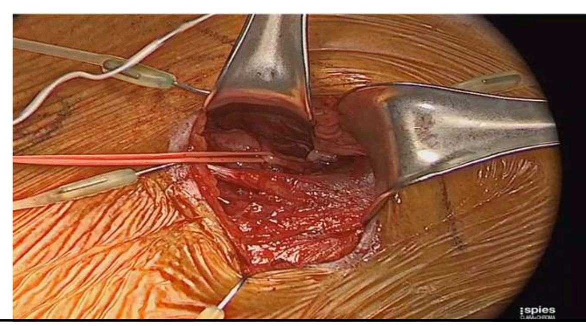
Target Muscles - Genioglossus, Geniohyoid

100



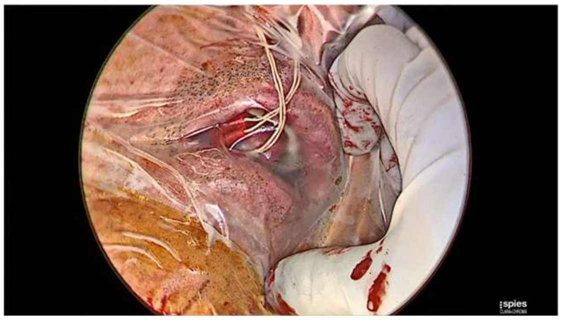
101

Upper Airway Stimulation for OSA



102

Tongue Movement Confirmation



103

104

CONCLUSIONS

01 Dentists play a role for obstructive sleep apnea care

02 Integrated, patient-centered customized patient Care considering ages, growth, preference and genetics

03 Start early, optimize the growth trajectory

105

CONCLUSIONS

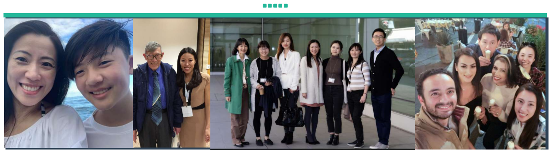
04 More clinical trials are necessary for accuracy of phenotyping, targeted treatment

05 Sleep Apnea pathophysiology is multifactorial. It is important to work with other providers for multidisciplinary collaborative care.

06 Never stop smiling ☺ !

106

Thank You



Stanford MEDICINE

www.draudreyyoon.com

www.smilesbydryoon.com

audrey12@stanford.edu jungdds@gmail.com https://www.facebook.com/audrey.yoon

Audrey Yoon, DDS, MS

107